

Jatene's Arterial Repair for Transposition of the Great Vessels

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In 1975, Adib Jatene demonstrated the clinical feasibility of an arterial repair for transposition of the great vessels. This repair consisted of removing the left and right coronary arteries and reanastomosing them to the adjacent pulmonary artery, which was to become the neo-aorta, and by transecting and transposing the aorta and main pulmonary artery. This repair varied from previous "arterial switch" procedures in that the aorta and pulmonary artery are divided distal from the aortic and pulmonary valves, and the coronary arteries are

removed with a button of aortic wall facilitating the anastomosis to the neo-aorta. Initially limited to patients with transposition and a large ventricular septal defect "preparing" the left ventricle, the indications for the arterial repair have been expanded to include neonates with simple transposition and other more complex lesions. Early results suggest that this anatomical and physiological approach will eliminate many of the complications associated with the physiological atrial repairs.

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Anatomic correction of transposition of the great vessels

We present a new approach for anatomic correction of transposition of the great arteries. The two coronary arteries, with a piece of the aortic wall attached, are transposed to the posterior artery. The two aortic openings are closed with a patch. The aorta and pulmonary artery are transected, contraposed, and then anastomosed. The interventricular septal defect is closed with a patch, through a right ventriculotomy approach, because the right ventricle is no longer part of the systemic circulation. Two patients, aged 3 months and 40 days and weighing 4,200 and 3,700 grams, respectively, were operated upon with deep hypothermia and total circulatory arrest. There was good recovery from the operation, with normal cardiocirculatory conditions. Renal failure developed in the first patient, and she died on the third postoperative day. During this time the cardiocirculatory conditions were good. The second patient made an uneventful recovery. Hemodynamic studies 20 days after the operation showed complete correction of the malformation. Five and one-half months after the operation, he weighs 7,500 grams, and his development is very good. We believe that this operation will be reproducible by most cardiovascular surgeons and will be an alternative to the Mustard procedure, especially for those patients with interventricular septal defect and pulmonary hypertension.

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The arterial correction of transposition of the great arteries became a reality with Adib Jatene's repair reported in 1975 [1] and 1976 [2]. As is often the case, this anatomical as well as physiological repair is striking in its

elegant simplicity and straightforward solution, which had eluded earlier surgical investigators.

Historical Considerations

Bailey and associates [3] attempted transposing the great vessels employing circulatory arrest in 1953. With no

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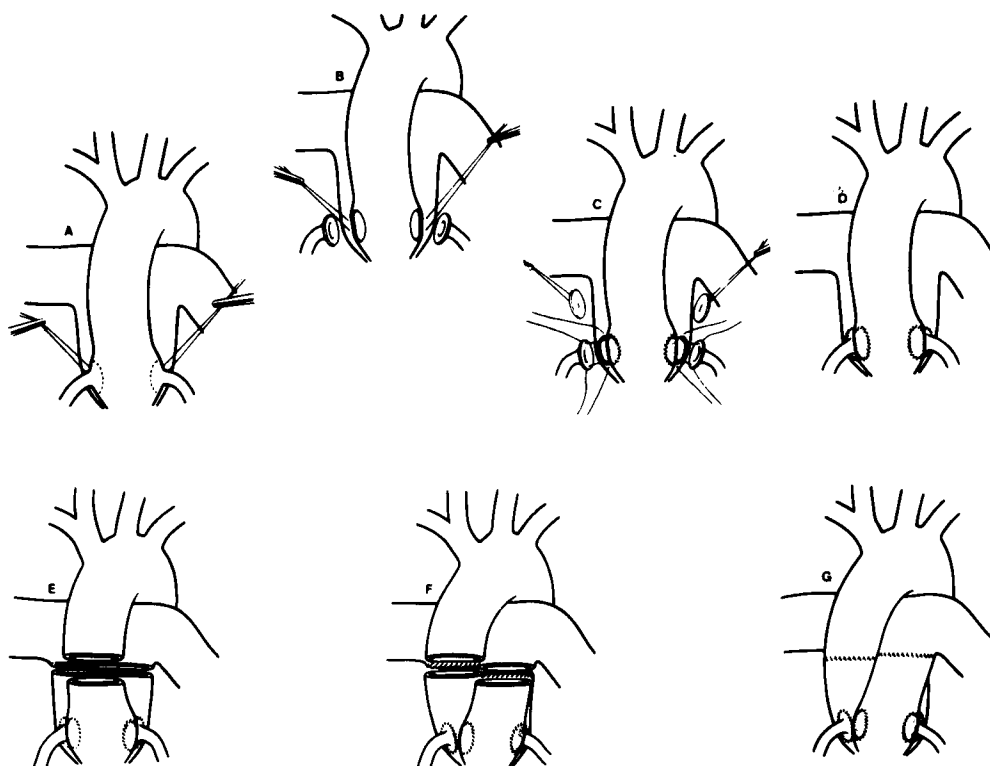


Fig 1. Jatene's original illustration demonstrating the anatomical correction employing aortic buttons to facilitate reanastomosis of the coronary arteries to the neo-aorta. (Reproduced from Jatene AD, Fontes VF, Paulista PP, et al. Anatomic correction of transposition of the great vessels. *J Thorac Cardiovasc Surg* 1976;72:364-70, with permission of Mosby-Year Book, Inc.)

survivors, and concern with the neopulmonary position of the coronary arteries, the investigators abandoned the procedure.

In 1954, Mustard and co-workers [4] attempted an arterial repair in 7 patients. The repair incorporated the left coronary artery but not the right coronary artery in the neo-aorta. Uniform failure was secondary to coronary distribution forcing abandonment of the procedure, low cardiac output, and problems associated with early cardiopulmonary bypass employing a monkey-lung oxygenator. Mustard's later attempts centered on the arterial repair.

The next year, Senning [5] attempted an arterial switch procedure in which the aorta and pulmonary artery were transected just above the aortic and pulmonary valves, respectively, and transposed, moving both the coronary arteries en bloc with the aorta. Having no clinical success with this procedure, Senning abandoned it in 1957 for the arterial repair bearing his name.

Other suggested arterial repairs [6-9] have been offered either theoretically or in animal models. The overwhelming dilemmas facing these earlier investigators have been the coronary arteries and the physiologically unprepared left ventricle in simple transposition, as the pulmonary resistance and hence pressure dropped after birth.

Jatene's Arterial Repair

After these early attempts at arterial repair for transposition of the great arteries, most efforts focused upon atrial

repairs which enjoyed greater success. In 1975 in a preliminary report [1] and in 1976 in the United States [2], Jatene and associates reported their clinical experience with an arterial repair. Although the first attempt was unsuccessful secondary to renal failure in the early post-operative period, the second attempt on a 40-day-old, 3,700-g infant with transposition and a large ventricular septal defect was successful. The repair (Fig 1), employing hypothermic circulatory arrest, consisted of removing the left and right coronary arteries and reanastomosing them to the adjacent pulmonary artery, which was to become the neo-aorta, and transecting and transposing the aorta and main pulmonary artery. The ventricular septal defect was closed with a Dacron patch through a right ventriculotomy.

This repair varied from previous procedures in two ways. First, the aorta and pulmonary artery are divided distally, far from the aortic and pulmonary valves, thus avoiding coronary artery and valve distortion. Second, the coronary arteries are removed with a button of aortic wall facilitating the anastomosis to the neo-aorta, and minimizing both the early and late risk of ostial stenosis.

The Role of the Arterial Repair Today

After the early experience of Jatene and others, attention was turned to modifications of the surgical technique and expanding the indications of the arterial repair. Although Jatene's repair initially used no graft material in the aortic or pulmonary anastomosis, it became apparent that the

pulmonary anastomosis was often under tension, inducing obstruction. Conduits and graft material were used to prevent this but raised concerns of long-term results [10]. Lecompte and associates [11], in 1981, introduced the maneuver of placing the aorta behind the dissected pulmonary artery, thus reducing tension and eliminating the need for a prosthetic graft. Variations of the coronary artery anatomy including a single coronary os were initially contraindications to the arterial switch repair. Modifications of the repair have been devised so that all but a very few coronary patterns can be successfully repaired [12-14].

Jatene noted early in his experience that the unprepared left ventricle was not capable of performing under systemic pressure conditions after an anatomical repair [15]. The arterial switch procedure was thus limited to anatomical variations where the left ventricular pressure was already near or equal to systemic pressure such as in patients with large ventricular septal defects or repairable pulmonary stenosis. Yacoub and colleagues [16] introduced the two-stage repair for simple transposition in which the pulmonary artery is banded in the first stage to prepare the left ventricle and permit an arterial switch as a second procedure. The concept of neonatal arterial repair [14, 17] has expanded the criteria to nearly all patients with simple transposition. The repair is performed within the first few weeks of life while the left ventricle remains prepared by the physiological pulmonary hypertension of the newborn in transition from fetal to mature pulmonary hemodynamics.

Recent experience in a large cooperative study [18] has demonstrated that the arterial repair in the neonate can be performed by an experienced surgical team with a low mortality rate. Intermediate-term results are promising despite some late complications such as valvar insufficiency and pulmonary stenosis in a small number of children. It is to be hoped that the late complications of arterial repairs, including arrhythmias, baffle obstruction, tricuspid regurgitation, and right ventricular dysfunction, can be avoided with the Jatene repair and its modifications.

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Editor's Note

We asked Dr Jatene to comment on Dr Dunn's article.

Invited Commentary

I feel honored by Dr Dunn, who wrote this paper, and by Dr Ferguson for asking me to add a comment.

It is not easy for a South American surgeon to make

new contributions, because most of us work under difficult conditions, which preclude investigations and research. However, I had the good fortune of being trained at the University Hospital by Professor Zerbini, with whom I worked from 1951, when I was still a medical student, to 1962, when I left to start my own group. Even after leaving I still remained in close contact with him. Together with Luiz Décourt, Dante Pazzanese, and a